Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	*(a)*		Award 1 mark for ± 0.1 [cm] **or** measure to resolution of ruler Award 2 marks for ± 0.2 [cm]	1	1		2	1	2
	(b)	(i)	$V = \pi\left(\frac{d}{2}\right)^2 l = \pi\left(\frac{1.5\times10^{-3}}{2}\right)^2 \times 11.5\times10^{-2}$ (subst and convincing change of units for l and d) (1) = 2.03×10^{-7} m^3 **or** 0.2 cm^3 **or** 203mm^3 (1) **unit mark**	1	1		2	2	2
		(ii)	$p_V = 2p_d + p_l = 2\frac{0.1}{1.5}(1) + \frac{0.2}{11.5}(1)$ (or by impl.) = 0.133 + 0.017 = 0.150 (1) allow **ecf** for this mark, on the use of 0.1 instead of 0.2 only Hence p_V = 15 % (**Alternative:** use percentages throughout)	1	1 1		3	3	3
	(c)	(i)	From intercepts with x-axis, mean = – 270 [°C] (1) Uncertainty = 20 [°C] (1) Award 1 mark only if 270 used		2		2	1	2
		(ii)	**Any 4 ×(1) from:** -Straight line -Intercept is consistent i.e. –273 °C / line would go through the origin if temp plotted in K -Passes through all error bars -Volume linked to length -$V \propto T$ **or** $l \propto T$			4	4	1	4
	(d)		Kinetic or internal energy (or velocity) approaches minimum / zero. Accept very little KE / stopped moving / molecules stop / stops vibrating Don't accept KE decreases greatly / superconduct or superfluid	1			1		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(e)		**Any 2 ×(1) from:** -Meniscus or equivalent linked to liquid pellet -Moving readings -Expansion of glass -Gas not ideal -Variation in atmospheric pressure -Gas and liquid at different temperatures -Friction / viscosity of liquid pellet -Parallax / looking at eye level -Ruler not parallel to tube don't accept just ruler vertical Accept inaccuracy of thermometer Don't accept resolution of thermometer			2	2		2
			Question 1 total	**4**	**6**	**6**	**16**	**8**	**15**